#1 【学術論文】 The design of soft-actuated upper-body exoskeletons for movement assistance and physical therapy

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1. 論文・ニュース概要

None

2. 公開ステータス

- オープンアクセス区分: オープンアクセス (全文閲覧可能)
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3. 英文アブストラクト (原文抄録)

This thesis presents the design, implementation, andvalidationof novel soft-actuatedexoskeletons formovementassistanceandrehabilita tion.Motivatedbytheneedtoenhancequalityoflifeforindividualswith neurologicalimpairments,suchasstrokesurvivorsandspinal-cord-injured individuals, thisworkexploreshybridassistivedevices thatmerge the comfort,compliance,andweightofsoftactuatorswiththestability,relia bility,andalignmentprovidedbyrigidstructures.Twokeysystemswere introduced: (1)ahighdegree-of-freedom(DOF) soft-actuatedarmand spineexoskeletonthatuseslightweightpolyethylenepneumaticactuators integrated intoa3D-printed frame, achievingover sevenDOFs includ ingforearmandshouldersupination/pronationmovements,easilycarried andmobile, capableof 3-dimensionalmovement. Experimental evalua tionsdemonstrate that theorigami-inspiredsystemdelivers torquesup to4.46Nm, supports full 3Darticulation, andeffectivelyredistributes loadthroughasegmentedelasticspine, enablingwearablegravitycom pensationevenforuserswithlimitedstrength. (2)elastic-tensegrityjoints inspiredbybiologicaltensegritytounlockmulti-axisflexibilitywhilepre servingmechanical reliabilityanduser comfort. Byunlocking the full rangeof soft-actuatormotionandeliminating the rigidityand friction of traditional joints, thesedesigns substantiallyenhance comfort, flex ibility, andpassivestabilitythatdemonstrate intelligentphysicaldesign canoutperformcontrol-basedcompensation.Theelastic-tensegrityjoints, whencombinedwithelectromyography(EMG)andelectroencephalogra phy(EEG) control, achieve smooth, adaptivemotion, passivedamping of tremors, andaccurate intentdetection(EMGaccuracy100%; EEG upto82%, intentpredictionis improvedbyzero-crossingmethods),val idatingthe feasibilityofbi-manual rehabilitationandbrain-machine in terfacingwithout theelectromagnetic interferenceassociatedwithrigid actuators.Beyondhardwareinnovations,supportingstudiesincludemul timodal feedbackexperimentsthatenhancethesenseofownership(SoO) overvirtuallimbs,suggestingsynergisticpotentialforimmersivetherapies andaligningwiththebroaderthemeofreconnectingthebrain-bodyloop. Together,thesecontributionsaddressimportantgapsincurrentassistive technologiesandthehuman-machineinterface(HMI);affordability,acces sibility,comfort,andadaptability,aswellasopeningupanewwholenew generationofexoskeletondesignsforclinicalandat-homerehabilitation.

4. 原文確認のご案内

研究の詳細データや査読全文については、必ず上記原文URLよりご確認ください。

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